through the sudden change in the gravitational force governing its motion.

This conclusion is in direct conflict with special relativity, since the latter ensures that no information can be transmitted faster than the speed of light—instantaneous transmission violates this precept maximally.

In the early part of the twentieth century, therefore, Einstein realized that the tremendously successful Newtonian theory of gravity was in conflict with his special theory of relativity. Confident in the veracity of special relativity and notwithstanding the mountain of experimental support for Newton's theory, Einstein sought a new theory of gravity compatible with special relativity. This ultimately led him to the discovery of general relativity, in which the character of space and time again went through a remarkable transformation.

Einstein's Happiest Thought

Even before the discovery of special relativity, Newton's theory of gravity was lacking in one important respect. Although it can be used to make highly accurate predictions about how objects will move under the influence of gravity, it offers no insight into what gravity is. That is, how is it that two bodies that are physically separate from another, possibly hundreds of millions of miles apart if not more, nonetheless influence each other's motion? By what means does gravity execute its mission? This is a problem of which Newton himself was well aware. In his own words,

It is inconceivable, that inanimate brute matter, should, without the mediation of something else, which is not material, operate upon and affect other matter without mutual contact. That Gravity should be innate, inherent and essential to matter so that one body may act upon another at a distance thro' a vacuum without the mediation of anything else, by and through which their action and force may be conveyed, from one to another, is to me so great an absurdity that I believe no Man who has in philosophical matters a competent faculty of thinking can ever fall into it. Gravity must be caused by an agent acting constantly according to certain laws; but whether this agent be material or immaterial, I have left to the consideration of my readers.1

That is, Newton accepted the existence of gravity and went on to develop equations that accurately describe its effects, but he never